Power Query Tutorial: Health Care Analysis Project

In this tutorial, we will focus on a dataset from the health care sector. You will perform various data profiling, cleaning, and transformation tasks using the Power Query editor in Power BI. This example will cover working with patient data, medical visits, treatments, and hospital departments. The hands-on tutorial is designed to take approximately 1 hour to complete.

CSV Files:

You will work with the following files:

- Patients.csv
- Visits.csv
- Treatments.csv
- Departments.csv

Step-by-Step Tutorial

Make sure you follow the best practice of creating a staging layer and using Reference to create a copy and perform the following operations on that table.

1. Patients Table:

Promote Headers:

- Ensure headers are correctly promoted.

Data Types:

- Set patient_id as Whole Number.
- Set patient_name, patient_gender, and patient_zipcode as Text.
- Set birthdate as Date.

Conditional Column:

- Create a column age_group with values "Child" if age < 18, "Adult" if age >= 18 and < 65, otherwise "Senior".

Remove Duplicates:

- Remove duplicate rows based on patient_id.

Correct Errors:

- Replace any null values in patient_zipcode with "00000".

Extract Data from Text Columns:

- Extract the first three characters from the patient_zipcode column into a new column named zipcode_prefix.

Generate New Column Based on Mathematical Formula:

- Create a new column age by calculating the difference between the current year and the birth_year.

2. Visits Table:

Promote Headers:

- Ensure headers are correctly promoted.

Data Types:

- Set visit_id, patient_id, and department_id as Whole Number.
- Set visit_date as Date.
- Set visit_reason and visit_outcome as Text.

Extract Month:

- Add a new column visit_month by extracting the month from the visit_date column, formatted as Text.

Add Conditional Column:

- Create a column visit_season with values "Winter" if the month is Dec-Feb, "Spring" if Mar-May, "Summer" if Jun-Aug, otherwise "Fall".

Extract Data from Text Columns:

- Extract the day from the visit_date column and create a new column named visit_day.

Data Profiling:

- Use Column Profile to view the distribution and distinct count of visit_reason.

Remove Errors:

- Identify and remove rows where visit_date is in the future.

3. Treatments Table:

Promote Headers:

- Ensure headers are correctly promoted.

Data Types:

- Set treatment_id and visit_id as Whole Number.
- Set treatment_date as Date.
- Set treatment_cost as Decimal Number.

Data Profiling:

- Use Column Distribution to identify and replace any null values in treatment_type with "Unknown".

Group By:

- Group by treatment_type to calculate the average treatment_cost and rename it to Avg Treatment Cost.
- Delete the previous step

Identify and Correct Errors:

- Identify rows where treatment_cost is negative and replace with zero.

Extract Data from Text Columns:

- Extract the month from the treatment_date column and create a new column named treatment_month.

Generate New Column Based on Mathematical Formula:

- Create a new column treatment_total_cost by multiplying treatment_cost by 1.05 (to add a 5% tax).

4. Departments Table:

Promote Headers:

- Ensure headers are correctly promoted.

Data Types:

- Set department_id as Whole Number.
- Set department_name and department_head as Text.

Merge Columns:

- Add a column department_info by merging department_name and department_head with a comma and space separator.

Conditional Column:

- Create a column department_type with values "Clinical" if department_name contains "Clinic" or "Surgery", otherwise "Non-Clinical".

Remove Duplicates:

- Remove duplicate rows based on department_id.

Extract Data from Text Columns:

- Extract the first word from the department_name column into a new column named department_short_name.